

Grade 7 Mathematics — Mock Exam — Paper A

MEDIUM • NON-CALCULATOR • 75 MINUTES • 80 MARKS

NAME

DATE

SCORE / 80

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to work without one.
- 75 minutes for 80 marks — about a minute a mark. If a 3-mark question has taken you five minutes, leave it and come back to it.
- No formulas and no methods are given anywhere on this paper — not one. Remembering the right one is part of what is being tested.
- Give every fraction and every ratio in its simplest form, and convert to a common unit before you simplify.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – QUICK FIRE

1. Write down the answer to each of these.

(a) Write 84 as a product of its prime factors, in index form. [1]

(b) -7×-6 [1]

(c) $\sqrt[3]{216}$ [1]

(d) Write 3^4 as an ordinary number. [1]

(e) 45×0.1 [1]

2. Write down the answer to each of these.

(a) Round 6.472 to 1 decimal place. [1]

(b) 0.3×0.4 [1]

(c) Write $\frac{3}{8}$ as a decimal. [1]

(d) Write the ratio 18 : 24 in its simplest form. [1]

(e) Expand $3(x + 5)$. [1]

SECTION 2 – SHORT ANSWERS

3. Two numbers are 12 and 18.

(a) Find the highest common factor (HCF) of 12 and 18. [1]

(b) Find the lowest common multiple (LCM) of 12 and 18. [1]

One factor tree for each number answers both parts. Do not build them twice.

4. Simplify $(2^5 \times 2^3) \div 2^6$. Give your answer as an ordinary number. [2]

5. Work out:

(a) $6.2 \div 0.1$

[1]

(b) 6.2×0.01

[1]

One of these answers is bigger than 6.2 and one is smaller. Decide which before you write anything.

6. Write these decimals in ascending order — smallest first.

[2]

0.7 0.68 0.702 0.71

7. Write these fractions in ascending order — smallest first. You must show your working.

[2]

$\frac{2}{3}$ $\frac{5}{8}$ $\frac{7}{12}$ $\frac{3}{4}$

Two of these four are close together. Looking at them will not separate them.

8. Work out $4\frac{1}{4} - 1\frac{5}{6}$. Give your answer as a mixed number in its simplest form.

[2]

9. A pencil costs ₹p. A pen costs ₹3 more than a pencil. Write an expression, in its simplest form, for the total cost of 4 pencils and 2 pens.

[2]

10. Factorise fully:

(a) $6x + 15$

[1]

(b) $8y^2 - 12y$

[1]

“Fully” means nothing is left inside the bracket that both terms still share.

SECTION 3 – MULTI-STEP

11. Work out:

(a) -6×-4

[1]

(b) $-6 - 4$

[1]

(c) $-48 \div 8$

[1]

(a) and (b) look almost the same and are not. Two minus signs multiplied make a plus; two minus signs added do not.

12. Work out $4.68 \div 0.12$, showing your method.

[3]

13. Work out $6 \times 2\frac{3}{4}$. Give your answer as a mixed number.

[3]

14. Work out $8 \div \frac{2}{5}$. [3]

Dividing by a fraction smaller than 1 gives an answer bigger than 8.

15. The cost, ₹C, of hiring a hall for n hours is given by $C = 40n + 250$.
(a) Find C when $n = 12$. [2]

(b) Find n when $C = 1050$. [1]

16. Expand and simplify $5(2x - 3) - 2(x - 4)$. [3]

-2×-4 is plus 8. That one sign is where this question is usually lost.

17. Solve $5x + 7 = 3x + 19$. Show every step. [3]

18. A jacket costs ₹840. Its price is increased by 15%. Work out the new price. [3]

SECTION 4 – PROBLEMS

19. ₹3600 is shared between Asha and Bela in the ratio 3 : 5.

(a) Work out how much each of them receives. [2]

(b) Bela's share is then increased by 20%. Work out her new amount. [2]

(c) How much more than Asha's share is Bela's new amount? [1]

Check your two answers to (a) add back up to 3600 before you go on.

20. Ravi thinks of a number, n . He multiplies it by 4 and then subtracts 7. His answer is the same as adding 11 to the number he first thought of.

(a) Form an equation in n . [2]

(b) Solve your equation, and check your answer works. [3]

21. Two shops sell the same rice.
Shop A: 1.5 kg for ₹96. Shop B: 2.5 kg for ₹155.

(a) Work out the price per kilogram at each shop. [3]

(b) Write down which shop is better value. [1]

(c) Work out the cost of 6 kg at the better-value shop. [1]

22. Two lights are switched on together at 9:00:00. One flashes every 12 seconds and the other flashes every 18 seconds.

(a) Work out how many seconds pass before they next flash together. [3]

(b) Write down the time at which they next flash together. [1]

(c) How many times do they flash together in the 6 minutes after 9:00:00? [1]

23. A shirt costs ₹800. In one week its price is increased by 25%. The following week the new price is reduced by 20%.

(a) Work out the price after the increase. [2]

(b) Work out the price after the reduction. [2]

(c) Write down the overall percentage change from ₹800. [1]

24. A rectangle has length $(x + 4)$ cm and width 3 cm.

(a) Write an expression for its area. Expand your answer. [2]

(b) Write an expression for its perimeter, in its simplest form. [2]

(c) Factorise your expression for the perimeter. [1]

End of paper. Before you hand it in, check every answer that came out as a round, comfortable number — those are the ones a wrong method produces most often.